Assignment 2 Systematic Review (Part A)

Mohammad Kaosain Akbar

UCID: 30136155

Topic of Research: A Systematic Literature Review of Artificial Intelligence in Software Release Engineering

The topic of research which I have selected is examining and exploring various Artificial Intelligence (AI) approaches used across various components of Software Release Engineering. This topic falls under Release Engineering (RE) which is an important area of Software Engineering (SE). SE in general focuses on development, testing, maintaining and management of software systems. RE on the other hand focuses on the process of delivering any software changes from developers to the end users. RE is comprised of six major components which includes Integration, Continuous Integration, Build Systems, Infrastructure-as-Code, Deployment and Release. These six components together ensure that software changes are properly integrated, built, tested and delivered to users.

With advancement in AI, researchers are proposing various Machine Learning (ML), Deep Learning (DL), Reinforcement Learning (RL) and Large Language Model (LLM) approaches to automate, improve and fix various SE activities. This systematic literature review (SLR) aims to investigate the application of these AI techniques across six major components of modern release engineering. I will examine the AI techniques being used, the challenges in using them and their effectiveness compared to traditional approaches in terms of efficiency, reliability and performance. I will also try to identify the limitations and research gaps in existing works.

The tentative research questions for the SLR include but not limited to:

1) What AI approaches have been addressed to the components of RE ?
2) How these AI approaches perform against non-AI baselines ?
3) What limitations and research gaps remain in using AI for different RE components ?

Since this SLR aims to explore AI research conducted across the six major RE components, there are multiple search terms associated with each component, as well as different AI approaches that can be used to find relevant research articles from various academic databases. All search strings used to identify relevant articles will be provided in the final version of the paper. Two examples of a proposed Google Scholar search string for finding research articles related to this SLR is given below.

- *("artificial intelligence" OR "machine learning" OR "deep learning" OR "large language model" OR "generative AI") AND ("merge conflict" OR "continuous integration" OR "build system" OR "infrastructure as code" OR "software deployment" OR "software release")*

- *("artificial intelligence" OR "machine learning"OR "deep learning" OR "reinforcement learning" OR "large language model*" OR "generative AI" OR "neural network*" OR "data mining" OR "intelligent agent*") AND ("merge conflict*" OR "merge resolution"OR "code review" OR "pull request*" OR "code integration" OR "branch merging" OR "merge prediction")*